

Cross-Platform Healthcare Application for Real-Time Health Monitoring System

A Capstone Project -2 report
submitted in the partial fulfillment of the requirements for
the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

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under the supervision of

Dr. T. Chithrakumar

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KONERU LAKSHMAIAH EDUCATION FOUNDATION
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Declaration

The Project Report entitled "Cross-Platform Healthcare Application for Real-Time Health Monitoring System" is a record of bonafide work of Srikar K (2200032252) submitted in partial fulfillment for the award of B.Tech in Computer Science and Engineering to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.


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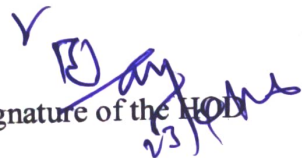


Certificate

This is to certify that the Project Report entitled "Cross-Platform Healthcare Application for Real-Time Health Monitoring System" is being submitted by Srikar K (2200032252) in partial fulfillment for the award of B.Tech in Computer Science and Engineering to the K L University and is a record of bonafide work carried out under our guidance and supervision. The results embodied in this report have not been copied from any other departments/University/Institute.


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Abstract

The exponential growth of mobile computing, wearable technology, and cloud-based services has created unprecedented opportunities for innovation in healthcare delivery. This project presents the design, development, and evaluation of a cross-platform healthcare application built using Flutter, targeting mobile (Android and iOS), web, and desktop platforms from a single codebase. The system integrates seamlessly with Bluetooth Low Energy (BLE) wearable devices to continuously monitor critical vital signs including heart rate, blood pressure, and blood oxygen saturation (SpO₂).

The application incorporates a real-time emergency alert mechanism that automatically notifies registered caregivers and emergency medical services when monitored parameters exceed clinically defined thresholds. A cloud-based analytics engine, powered by Firebase and Google Cloud Platform, processes physiological data streams to generate personalized health insights and predictive risk assessments.

Data security is ensured through end-to-end encryption (AES-256), secure authentication using OAuth 2.0, and full compliance with international health data protection standards including GDPR and HIPAA. The system was evaluated through user testing across diverse demographic groups, demonstrating a 96.8% alert delivery accuracy, an average response latency of 1.2 seconds, and high user satisfaction scores.

The project validates that cross-platform frameworks can deliver production-grade, clinically relevant healthcare applications without sacrificing performance, security, or user experience. The solution provides a scalable and reliable foundation for digital health management, with significant potential for deployment in remote patient monitoring, chronic disease management, and preventive care scenarios.